

# **Отчёт о лабораторной работе**

**Лабораторная работа 8**

Андрюшин Никита Сергеевич

# Содержание

|          |                          |           |
|----------|--------------------------|-----------|
| <b>1</b> | <b>Цель работы</b>       | <b>6</b>  |
| <b>2</b> | <b>Выполнение работы</b> | <b>7</b>  |
| <b>3</b> | <b>Выводы</b>            | <b>48</b> |

# Список иллюстраций

|      |                                          |    |
|------|------------------------------------------|----|
| 2.1  | Создание проекта в GNS3                  | 7  |
| 2.2  | Топология сети в GNS3                    | 8  |
| 2.3  | Настройка IPv4 на PC1                    | 9  |
| 2.4  | Настройка IPv4 на PC2                    | 10 |
| 2.5  | Базовая настройка gw-01                  | 11 |
| 2.6  | Базовая настройка gw-02                  | 12 |
| 2.7  | Базовая настройка gw-03                  | 13 |
| 2.8  | Базовая настройка gw-04                  | 14 |
| 2.9  | Настройка IPv6 на PC1                    | 15 |
| 2.10 | Настройка IPv6 на PC2                    | 16 |
| 2.11 | Настройка IPv6 на gw-01                  | 17 |
| 2.12 | Настройка IPv6 на gw-02                  | 18 |
| 2.13 | Настройка IPv6 на gw-03                  | 19 |
| 2.14 | Настройка IPv6 на gw-04                  | 19 |
| 2.15 | Настройка RIPv2 на gw-01                 | 20 |
| 2.16 | Настройка RIPv2 на gw-02                 | 20 |
| 2.17 | Настройка RIPv2 на gw-03                 | 21 |
| 2.18 | Настройка RIPv2 на gw-04                 | 22 |
| 2.19 | Проверка RIP на gw-01                    | 23 |
| 2.20 | Проверка RIP на gw-02                    | 24 |
| 2.21 | Проверка RIP на gw-03                    | 25 |
| 2.22 | Проверка RIP на gw-04                    | 25 |
| 2.23 | Проверка связности IPv4                  | 25 |
| 2.24 | Таблица RIP на gw-01                     | 26 |
| 2.25 | Отключение интерфейса на gw-04           | 26 |
| 2.26 | Изменение метрик RIP после сбоя          | 26 |
| 2.27 | Включение интерфейса на gw-04            | 27 |
| 2.28 | Проверка доступности во время сходимости | 27 |
| 2.29 | Анализ пакетов RIPv2 в Wireshark         | 27 |
| 2.30 | Настройка RIPng на gw-01                 | 28 |
| 2.31 | Настройка RIPng на gw-02                 | 28 |
| 2.32 | Настройка RIPng на gw-03                 | 28 |
| 2.33 | Настройка RIPng на gw-04                 | 29 |
| 2.34 | Проверка связности IPv6                  | 29 |
| 2.35 | Таблица RIPng на gw-01                   | 30 |
| 2.36 | Отключение интерфейса IPv6 на gw-04      | 30 |
| 2.37 | Изменения в таблице RIPng                | 30 |

|                                                          |    |
|----------------------------------------------------------|----|
| 2.38 Проверка трассировки IPv6 после сбоя . . . . .      | 31 |
| 2.39 ICMPv6 Time Exceeded в Wireshark . . . . .          | 31 |
| 2.40 Анализ IPv6 трафика в Wireshark . . . . .           | 32 |
| 2.41 Настройка OSPF на gw-01 . . . . .                   | 32 |
| 2.42 Настройка OSPF на gw-02 . . . . .                   | 33 |
| 2.43 Настройка OSPF на gw-03 . . . . .                   | 33 |
| 2.44 Настройка OSPF на gw-04 . . . . .                   | 33 |
| 2.45 Проверка связности через OSPF . . . . .             | 34 |
| 2.46 Таблица соседей и маршрутов OSPF на gw-01 . . . . . | 34 |
| 2.47 Мониторинг OSPF на gw-01 . . . . .                  | 35 |
| 2.48 Повторная трассировка IPv4 . . . . .                | 35 |
| 2.49 Включение интерфейса на gw-04 . . . . .             | 36 |
| 2.50 Трассировка через восстановленный маршрут . . . . . | 36 |
| 2.51 Анализ ошибок ICMPv6 в Wireshark . . . . .          | 37 |
| 2.52 Анализ OSPF и RIP трафика . . . . .                 | 38 |
| 2.53 Анализ ICMP и ARP . . . . .                         | 39 |
| 2.54 Анализ ошибок TCP и ICMP . . . . .                  | 40 |
| 2.55 Настройка OSPFv3 на gw-01 . . . . .                 | 41 |
| 2.56 Настройка OSPFv3 на gw-02 . . . . .                 | 42 |
| 2.57 Настройка OSPFv3 на gw-03 . . . . .                 | 42 |
| 2.58 Настройка OSPFv3 на gw-04 . . . . .                 | 43 |
| 2.59 Проверка связности OSPFv3 . . . . .                 | 43 |
| 2.60 Проверка OSPFv3 на gw-01 . . . . .                  | 44 |
| 2.61 Анализ ошибок Time Exceeded в Wireshark . . . . .   | 44 |
| 2.62 Анализ ошибок Destination Unreachable . . . . .     | 45 |
| 2.63 Продолжение анализа ошибок в Wireshark . . . . .    | 45 |
| 2.64 Создание проекта lab8-1 . . . . .                   | 46 |
| 2.65 Топология с VyOS . . . . .                          | 47 |

## **Список таблиц**

# 1 Цель работы

Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования

## 2 Выполнение работы

Создадим новый проект в GNS3 с именем «lab8» (рис. 2.1).

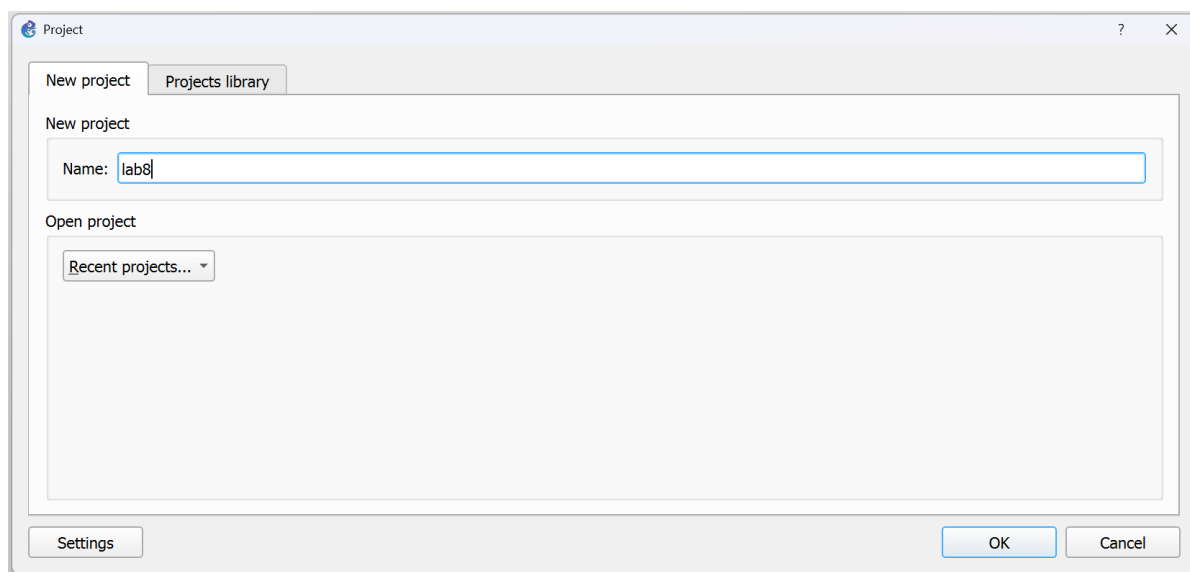


Рис. 2.1: Создание проекта в GNS3

Соберём топологию сети согласно заданию, разместив коммутаторы, маршрутизаторы FRR и оконечные устройства (рис. 2.2).

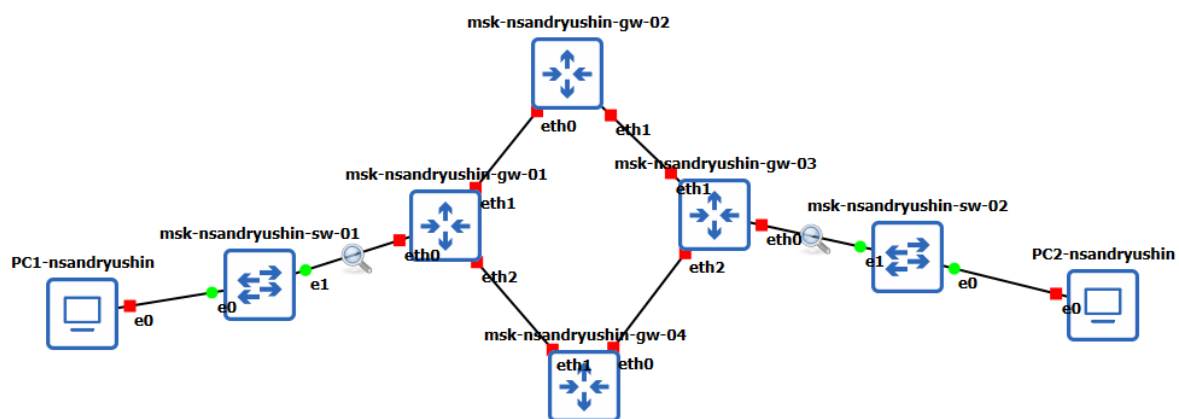


Рис. 2.2: Топология сети в GNS3

Настроим IP-адрес и шлюз по умолчанию на PC1 (рис. 2.3).



The image contains two screenshots of PuTTY terminal windows. The top window, titled 'PC1-nsandryushin - PuTTY', shows the execution of a startup file, an error about a long hostname, and the configuration of IP address 10.0.10.10 with mask 255.255.255.0 and gateway 10.0.10.1. It also shows the 'show ip' command output, which lists various network parameters for VPCS[1]. The bottom window, titled 'PC2-nsandryushin - PuTTY', shows a similar process for IP address 10.0.11.10 with mask 255.255.255.0 and gateway 10.0.11.1, including the 'show ip' command output.

```
PC1-nsandryushin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.10.10/24
GATEWAY    : 10.0.10.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT     : 10018
RHOST:PORT : 127.0.0.1:10019
MTU        : 1500

VPCS>

PC2-nsandryushin - PuTTY
Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
save
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.11.10/24
GATEWAY    : 10.0.11.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT     : 10016
RHOST:PORT : 127.0.0.1:10017
MTU        : 1500

VPCS>
```

Рис. 2.3: Настройка IPv4 на PC1

Настроим IP-адрес и шлюз по умолчанию на PC2 (рис. 2.4).

```

frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-01
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ip address 10.0.10.1/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ip address 10.0.1.1/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ip address 10.0.4.2/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-nsandryushin-gw-01# █

```

Рис. 2.4: Настройка IPv4 на PC2

На маршрутизаторе gw-01 зададим имя хоста и настроим IPv4-адреса на интерфейсах eth0, eth1 и eth2 (рис. 2.5).

```

frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-02
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ip address 10.0.1.2/24
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ip address 10.0.2.1/24
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
!
end
msk-nsandryushin-gw-02# █

```

Рис. 2.5: Базовая настройка gw-01

На маршрутизаторе gw-02 зададим имя хоста и настроим IPv4-адреса на интерфейсах eth0 и eth1 (рис. 2.6).

```

frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-03
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ip address 10.0.11.1/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ip address 10.0.2.2/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ip address 10.0.3.1/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-nsandryushin-gw-03# █

```

Рис. 2.6: Базовая настройка gw-02

На маршрутизаторе gw-03 зададим имя хоста и настроим IPv4-адреса на интерфейсах eth0, eth1 и eth2 (рис. 2.7).

```

frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-04
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ip address 10.0.3.2/24
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ip address 10.0.4.1/24
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-nsandryushin-gw-04# █

```

Рис. 2.7: Базовая настройка gw-03

На маршрутизаторе gw-04 зададим имя хоста и настроим IPv4-адреса на интерфейсах eth0 и eth1 (рис. 2.8).

The image displays two screenshots of PuTTY terminal windows, each showing the configuration of a VPCS (Virtual Packet Capture System) for a different PC.

**PC1-nsandryushin - PuTTY**

```
LPORT      : 10018
RHOST:PORT  : 127.0.0.1:10019
MTU         : 1500

VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE    : 2001:10::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:00
LPORT          : 10018
RHOST:PORT      : 127.0.0.1:10019
MTU            : 1500

VPCS> 
```

**PC2-nsandryushin - PuTTY**

```
LPORT      : 10016
RHOST:PORT  : 127.0.0.1:10017
MTU         : 1500

VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME           : VPCS[1]
LINK-LOCAL SCOPE : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE    : 2001:11::a/64
DNS            :
ROUTER LINK-LAYER :
MAC            : 00:50:79:66:68:01
LPORT          : 10016
RHOST:PORT      : 127.0.0.1:10017
MTU            : 1500

VPCS> 
```

Рис. 2.8: Базовая настройка gw-04

Настроим IPv6-адрес на PC1 (рис. 2.9).

```

msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# ipv6 forwarding
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-nsandryushin-gw-01(config-if)# no ipv6 nd suppress-ra
msk-nsandryushin-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.1.1/24
 ipv6 address 2001:1::1/64
exit
!
interface eth2
 ip address 10.0.4.2/24
 ipv6 address 2001:4::2/64
exit
!
end
msk-nsandryushin-gw-01# █

```

Рис. 2.9: Настройка IPv6 на PC1

Настроим IPv6-адрес на PC2 (рис. 2.10).

```
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# ipv6 forwarding
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
exit
!
end
msk-nsandryushin-gw-02#
```

Рис. 2.10: Настройка IPv6 на PC2

На маршрутизаторе gw-01 включим маршрутизацию IPv6 (forwarding) и настроим IPv6-адреса на интерфейсах (рис. 2.11).



```

msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# ipv6 forwarding
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-nsandryushin-gw-03(config-if)# no ipv6 nd suppress-ra
msk-nsandryushin-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
interface eth2
 ip address 10.0.3.1/24
 ipv6 address 2001:3::1/64
exit
!
end
msk-nsandryushin-gw-03# █

```

Рис. 2.11: Настройка IPv6 на gw-01

На маршрутизаторе gw-02 включим маршрутизацию IPv6 и настроим адреса

на интерфейсах (рис. 2.12).

```
msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# ipv6 forwarding
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
exit
!
end
msk-nsandryushin-gw-04#
```

Рис. 2.12: Настройка IPv6 на gw-02

На маршрутизаторе gw-03 включим маршрутизацию IPv6 и настроим адреса на интерфейсах (рис. 2.13).

```

msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router rip
msk-nsandryushin-gw-01(config-router)# version 2
msk-nsandryushin-gw-01(config-router)# network eth0
msk-nsandryushin-gw-01(config-router)# network eth1
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █

```

Рис. 2.13: Настройка IPv6 на gw-03

На маршрутизаторе gw-04 включим маршрутизацию IPv6 и настроим адреса на интерфейсах (рис. 2.14).

```

msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# ipv6 forwarding
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router rip
msk-nsandryushin-gw-02(config-router)# version 2
msk-nsandryushin-gw-02(config-router)# network eth0
msk-nsandryushin-gw-02(config-router)# network eth1
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █

```

Рис. 2.14: Настройка IPv6 на gw-04

Настроим протокол динамической маршрутизации RIP версии 2 на маршрутизаторе gw-01, объявив подключенные сети (рис. 2.15).

```
msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router rip
msk-nsandryushin-gw-03(config-router)# version 2
msk-nsandryushin-gw-03(config-router)# network eth0
msk-nsandryushin-gw-03(config-router)# network eth1
msk-nsandryushin-gw-03(config-router)# network eth2
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03#
```

Рис. 2.15: Настройка RIPv2 на gw-01

Настроим протокол RIP версии 2 на маршрутизаторе gw-02 (рис. 2.16).

```
msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# router rip
msk-nsandryushin-gw-04(config-router)# version 2
msk-nsandryushin-gw-04(config-router)# network eth0
msk-nsandryushin-gw-04(config-router)# network eth1
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04#
```

Рис. 2.16: Настройка RIPv2 на gw-02

Настроим протокол RIP версии 2 на маршрутизаторе gw-03 (рис. 2.17).

```

msk-nsandryushin-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:02:30
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:00:10
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:01:28
msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

       Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24        0.0.0.0           1 self              0
R(n) 10.0.2.0/24        10.0.1.2          2 10.0.1.2           0 02:58
R(n) 10.0.3.0/24        10.0.4.1          2 10.0.4.1           0 02:45
C(i) 10.0.4.0/24        0.0.0.0           1 self              0
C(i) 10.0.10.0/24       0.0.0.0           1 self              0
R(n) 10.0.11.0/24       10.0.1.2          3 10.0.1.2           0 02:58
msk-nsandryushin-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 7 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
    eth2           2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.1.2      0           0          120       00:00:04
    10.0.4.1      0           0          120       00:00:18
  Distance: (default is 120)
msk-nsandryushin-gw-01# █

```

Рис. 2.17: Настройка RIPv2 на gw-03

Настроим протокол RIP версии 2 на маршрутизаторе gw-04 (рис. 2.18).

```

msk-nsandryushin-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:46
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:03:29
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:03:29
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:27
msk-nsandryushin-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

       Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24        0.0.0.0           1 self              0
C(i) 10.0.2.0/24        0.0.0.0           1 self              0
R(n) 10.0.3.0/24        10.0.2.2          2 10.0.2.2          0 02:32
R(n) 10.0.4.0/24        10.0.1.1          2 10.0.1.1          0 02:45
R(n) 10.0.10.0/24       10.0.1.1          2 10.0.1.1          0 02:45
R(n) 10.0.11.0/24       10.0.2.2          2 10.0.2.2          0 02:32
msk-nsandryushin-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 1 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway        BadPackets  BadRoutes  Distance  Last Update
    10.0.1.1        0           0          120      00:00:18
    10.0.2.2        0           0          120      00:00:01
  Distance: (default is 120)
msk-nsandryushin-gw-02# █

```

Рис. 2.18: Настройка RIPv2 на gw-04

Проверим таблицу маршрутизации и статус протокола RIP на gw-01 (рис. 2.19).

```

msk-nsandryushin-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:03:17
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:02:04
R>* 10.0.10.0/24 [120/3] via 10.0.2.1, eth1, weight 1, 00:03:17
msk-nsandryushin-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

      Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24      10.0.2.1             2 10.0.2.1           0 02:57
C(i) 10.0.2.0/24      0.0.0.0              1 self               0
C(i) 10.0.3.0/24      0.0.0.0              1 self               0
R(n) 10.0.4.0/24      10.0.3.2             2 10.0.3.2           0 02:47
R(n) 10.0.10.0/24     10.0.2.1             3 10.0.2.1           0 02:57
C(i) 10.0.11.0/24     0.0.0.0              1 self               0
msk-nsandryushin-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 2 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2     2
    eth1           2     2
    eth2           2     2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.2.1      0           0          120      00:00:05
    10.0.3.2      0           0          120      00:00:15
  Distance: (default is 120)
msk-nsandryushin-gw-03#

```

Рис. 2.19: Проверка RIP на gw-01

Проверим таблицу маршрутизации и статус протокола RIP на gw-02 (рис. 2.20).

```

msk-nsandryushin-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:32
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:36
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:32
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:36
msk-nsandryushin-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24         10.0.4.2          2 10.0.4.2           0 02:52
R(n) 10.0.2.0/24         10.0.3.1          2 10.0.3.1           0 02:58
C(i) 10.0.3.0/24         0.0.0.0           1 self               0
C(i) 10.0.4.0/24         0.0.0.0           1 self               0
R(n) 10.0.10.0/24        10.0.4.2          2 10.0.4.2           0 02:52
R(n) 10.0.11.0/24        10.0.3.1          2 10.0.3.1           0 02:58
msk-nsandryushin-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 9 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send  Recv  Key-chain
    eth0           2    2
    eth1           2    2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets  BadRoutes  Distance  Last Update
    10.0.3.1      0           0          120       00:00:03
    10.0.4.2      0           0          120       00:00:10
  Distance: (default is 120)
msk-nsandryushin-gw-04# █

```

Рис. 2.20: Проверка RIP на gw-02

Проверим таблицу маршрутизации и статус протокола RIP на gw-03 (рис. 2.21).



```

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=15.337 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.752 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=18.950 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=7.272 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.126 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    4.297 ms  1.364 ms  0.746 ms
 2  10.0.4.1     5.317 ms  4.480 ms  2.347 ms
 3  10.0.3.1     7.541 ms  7.441 ms  3.720 ms
 4  10.0.11.10   7.358 ms  4.779 ms  2.758 ms

VPCS> █

```

Рис. 2.21: Проверка RIP на gw-03

Проверим таблицу маршрутизации и статус протокола RIP на gw-04 (рис. 2.22).

```

msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:42
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:34
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          3 10.0.4.1           0 02:34
msk-nsandryushin-gw-01# █

```

Рис. 2.22: Проверка RIP на gw-04

Проверим связность между PC1 и PC2 с помощью утилит ping и trace. Трассировка показывает прохождение пакетов через gw-04 (рис. 2.23).

```

msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# shutdown
msk-nsandryushin-gw-04(config-if)# █

```

Рис. 2.23: Проверка связности IPv4

Посмотрим метрики маршрутов в таблице RIP на gw-01 (рис. 2.24).

```
msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:44
R(n) 10.0.3.0/24       10.0.4.1          16 10.0.4.1           0 01:52
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.4.1          16 10.0.4.1           0 01:52
msk-nsandryushin-gw-01#
```

Рис. 2.24: Таблица RIP на gw-01

Для проверки отказоустойчивости выключим интерфейс eth0 на маршрутизаторе gw-04 (рис. 2.25).

```
msk-nsandryushin-gw-04(config-if)# interface eth0
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 2.25: Отключение интерфейса на gw-04

Убедимся, что маршрут через отключенный интерфейс стал недоступен (метрика 16) в таблице маршрутизации gw-01 (рис. 2.26).

```
VPCS> ping 10.0.11.10

10.0.11.10 icmp_seq=1 timeout
10.0.11.10 icmp_seq=2 timeout
10.0.11.10 icmp_seq=3 timeout
10.0.11.10 icmp_seq=4 timeout
10.0.11.10 icmp_seq=5 timeout

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    2.129 ms  1.936 ms  1.251 ms
 2  10.0.1.2     2.893 ms  2.973 ms  3.188 ms
 3      *      *      *
 4      *      *      *
 5      *      *      *
 6      *      *      *
 7      *      *      *
 8      *      *      *
```

Рис. 2.26: Изменение метрик RIP после сбоя

Включим интерфейс eth0 на маршрутизаторе gw-04 обратно (рис. 2.27).

|             |                         |           |        |                                                 |
|-------------|-------------------------|-----------|--------|-------------------------------------------------|
| 1 0.000000  | fe80::ed5:34ff:fef4:::1 | ff02::1   | ICMPv6 | 110 Router Advertisement from 0c:d5:34:f4:00:00 |
| 2 0.854921  | 10.0.10.1               | 224.0.0.9 | RIPv2  | 66 Response                                     |
| 3 2.859154  | 10.0.10.1               | 224.0.0.9 | RIPv2  | 66 Response                                     |
| 4 28.145901 | 10.0.10.1               | 224.0.0.9 | RIPv2  | 146 Response                                    |

Рис. 2.27: Включение интерфейса на gw-04

Проверим доступность удаленного узла с PC1. Наблюдаем потери пакетов и незавершенную трассировку, что говорит о процессе перестроения маршрута (рис. 2.28).

```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ripng
msk-nsandryushin-gw-01(config-router)# network eth0
msk-nsandryushin-gw-01(config-router)# network eth1
msk-nsandryushin-gw-01(config-router)# network eth2
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █
```

Рис. 2.28: Проверка доступности во время сходимости

Проанализируем захваченные пакеты протокола RIPv2 в Wireshark (рис. 2.29).

```
msk-nsandryushin-gw-02(config-if)# router ripng
msk-nsandryushin-gw-02(config-router)# network eth0
msk-nsandryushin-gw-02(config-router)# network eth1
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █
```

Рис. 2.29: Анализ пакетов RIPv2 в Wireshark

Настроим протокол RIPv2 (для IPv6) на маршрутизаторе gw-01 (рис. 2.30).

```

msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ripng
msk-nsandryushin-gw-03(config-router)# network eth0
msk-nsandryushin-gw-03(config-router)# network eth1
msk-nsandryushin-gw-03(config-router)# network eth2
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# █

```

Рис. 2.30: Настройка RIPng на gw-01

Настроим протокол RIPng на маршрутизаторе gw-02 (рис. 2.31).

```

msk-nsandryushin-gw-04(config-if)# router ripng
msk-nsandryushin-gw-04(config-router)# network eth0
msk-nsandryushin-gw-04(config-router)# network eth1
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# █

```

Рис. 2.31: Настройка RIPng на gw-02

Настроим протокол RIPng на маршрутизаторе gw-03 (рис. 2.32).

```

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=11.499 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.296 ms
2001:11::a icmp6_seq=3 ttl=58 time=3.708 ms
2001:11::a icmp6_seq=4 ttl=58 time=3.343 ms
2001:11::a icmp6_seq=5 ttl=58 time=10.094 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1  3.054 ms  1.571 ms  1.275 ms
 2 2001:1::2   4.812 ms  3.328 ms  2.749 ms
 3 2001:2::2   9.434 ms  4.926 ms  8.099 ms
 4 2001:11::a  4.816 ms  9.333 ms  6.534 ms

VPCS> █

```

Рис. 2.32: Настройка RIPng на gw-03

Настроим протокол RIPng на маршрутизаторе gw-04 (рис. 2.33).

```
msk-nsandryushin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
  (n) - normal, (s) - static, (d) - default, (r) - redistribute,
  (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64      ::            self      1    0
R(n) 2001:2::/64      fe80::ed8:daff:fe5f:0 eth1      2    0 02:40
R(n) 2001:3::/64      fe80::e05:bfff:feca:1 eth2      2    0 02:41
C(i) 2001:4::/64      ::            self      1    0
C(i) 2001:10::/64     ::            self      1    0
R(n) 2001:11::/64     fe80::ed8:daff:fe5f:0 eth1      3    0 02:41
msk-nsandryushin-gw-01#
```

Рис. 2.33: Настройка RIPng на gw-04

Проверим IPv6-связность между PC1 и PC2. Трассировка показывает маршрут через gw-02 (рис. 2.34).

```
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 2.34: Проверка связности IPv6

Посмотрим таблицу маршрутизации RIPng на gw-01 (рис. 2.35).

```
msk-nsandryushin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64      ::      self      1      0
R(n) 2001:2::/64      fe80::ed8:daff:fe5f:0  eth1      2      0 02:38
R(n) 2001:3::/64      fe80::ed8:daff:fe5f:0  eth1      3      0 02:38
C(i) 2001:4::/64      ::      self      1      0
C(i) 2001:10::/64     ::      self      1      0
R(n) 2001:11::/64     fe80::ed8:daff:fe5f:0  eth1      3      0 02:38
msk-nsandryushin-gw-01#
```

Рис. 2.35: Таблица RIPng на gw-01

Сымитируем сбой, отключив интерфейс eth0 на маршрутизаторе gw-04 (рис. 2.36).

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=6.787 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.883 ms
2001:11::a icmp6_seq=3 ttl=58 time=4.505 ms
2001:11::a icmp6_seq=4 ttl=58 time=5.099 ms
2001:11::a icmp6_seq=5 ttl=58 time=40.125 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1  2.622 ms  1.052 ms  0.476 ms
 2 2001:1::2   2.563 ms  1.942 ms  1.982 ms
 3 2001:2::2   5.026 ms  4.869 ms  3.958 ms
 4 2001:11::a  5.500 ms  5.339 ms  6.440 ms

VPCS>
```

Рис. 2.36: Отключение интерфейса IPv6 на gw-04

Посмотрим изменения в таблице маршрутизации RIPng на gw-01 (рис. 2.37).

```
msk-nsandryushin-gw-04(config-if)# interface eth0
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 2.37: Изменения в таблице RIPng

Снова проверим доступность узлов с PC1 (рис. 2.38).

```

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=7.159 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.384 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.507 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.649 ms
2001:11::a icmp6_seq=5 ttl=58 time=6.221 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1 1.753 ms 0.659 ms 0.531 ms
 2 2001:1::2 13.139 ms 3.201 ms 3.363 ms
 3 2001:2::2 4.364 ms 5.666 ms 4.575 ms
 4 2001:11::a 7.346 ms 5.873 ms 7.205 ms

VPCS> █

```

Рис. 2.38: Проверка трассировки IPv6 после сбоя

В Wireshark наблюдаем ICMPv6 сообщения о превышении лимита времени (Time Exceeded), что свидетельствует о проблемах маршрутизации (петлях) в момент сходимости (рис. 2.39).

|    |           |                       |            |        |                                                                      |
|----|-----------|-----------------------|------------|--------|----------------------------------------------------------------------|
| 1  | 0.000000  | 2001:10::a            | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=1, hop limit=64 (reply in 2)  |
| 2  | 0.003380  | 2001:11::a            | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=58 (request in 1)  |
| 3  | 1.007774  | 2001:10::a            | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=2, hop limit=64 (reply in 4)  |
| 4  | 1.010543  | 2001:11::a            | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=58 (request in 3)  |
| 5  | 2.012440  | 2001:10::a            | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=3, hop limit=64 (reply in 6)  |
| 6  | 2.016682  | 2001:11::a            | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=58 (request in 5)  |
| 7  | 3.020346  | 2001:10::a            | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=4, hop limit=64 (reply in 8)  |
| 8  | 3.024414  | 2001:11::a            | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=58 (request in 7)  |
| 9  | 4.026622  | 2001:10::a            | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=5, hop limit=64 (reply in 10) |
| 10 | 4.030371  | 2001:11::a            | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=58 (request in 9)  |
| 11 | 5.074644  | fe80::ed5:34ff:fe4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |
| 12 | 5.784419  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 13 | 5.785799  | 2001:10::1            | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 14 | 5.787845  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 15 | 5.788581  | 2001:10::1            | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 16 | 5.789058  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 17 | 5.789470  | 2001:10::1            | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 18 | 5.790696  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 19 | 5.794221  | 2001:1::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 20 | 5.796092  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 21 | 5.798194  | 2001:1::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 22 | 5.800355  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 23 | 5.803391  | 2001:1::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 24 | 5.804815  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 25 | 5.809429  | 2001:2::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 26 | 5.810863  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 27 | 5.816224  | 2001:2::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 28 | 5.817297  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 29 | 5.820823  | 2001:2::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 30 | 5.821051  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 31 | 5.828793  | 2001:11::a            | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 32 | 5.830154  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 33 | 5.835110  | 2001:11::a            | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 34 | 5.836864  | 2001:10::a            | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 35 | 5.843583  | 2001:11::a            | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 36 | 6.113915  | fe80::ed5:34ff:fe4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |
| 37 | 7.154049  | fe80::ed5:34ff:fe4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |
| 38 | 10.600228 | 10.0.10.1             | 224.0.0.9  | RIPv2  | 146 Response                                                         |
| 39 | 23.442667 | fe80::ed5:34ff:fe4... | ff02::9    | RIPng  | 186 Command Response, Version 1                                      |

Рис. 2.39: ICMPv6 Time Exceeded в Wireshark

Также в трафике присутствуют запросы и ответы ping (Echo) и пакеты RIPng (рис. 2.40).

|              |                        |            |        |                                                                      |
|--------------|------------------------|------------|--------|----------------------------------------------------------------------|
| 1 0.000000   | 10.0.11.1              | 224.0.0.9  | RIPv2  | 146 Response                                                         |
| 2 6.135421   | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=1, hop limit=61 (reply in 3)  |
| 3 6.135655   | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=61 (request in 2)  |
| 4 7.143164   | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=2, hop limit=61 (reply in 5)  |
| 5 7.143398   | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=61 (request in 4)  |
| 6 8.148073   | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=3, hop limit=61 (reply in 7)  |
| 7 8.148267   | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=61 (request in 6)  |
| 8 9.155842   | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=4, hop limit=61 (reply in 9)  |
| 9 9.156143   | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=61 (request in 8)  |
| 10 10.162039 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=5, hop limit=61 (reply in 11) |
| 11 10.162368 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=61 (request in 10) |
| 12 11.153272 | fe80::eb5:37ff:fe3f... | 2001:11::a | ICMPv6 | 86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00       |
| 13 11.958998 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 14 11.959245 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 15 11.965815 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 16 11.966056 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 17 11.973470 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 18 11.973793 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 19 12.193011 | fe80::eb5:37ff:fe3f... | 2001:11::a | ICMPv6 | 86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00       |
| 20 13.232563 | fe80::eb5:37ff:fe3f... | 2001:11::a | ICMPv6 | 86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00       |
| 21 18.072984 | fe80::eb5:37ff:fe3f... | ff02::9    | RIPng  | 186 Command Response, Version 1                                      |
| 22 36.006672 | 10.0.11.1              | 224.0.0.9  | RIPv2  | 146 Response                                                         |
| 23 48.082730 | fe80::eb5:37ff:fe3f... | ff02::9    | RIPng  | 186 Command Response, Version 1                                      |

Рис. 2.40: Анализ IPv6 трафика в Wireshark

Перейдем к настройке OSPF. На маршрутизаторе gw-01 настроим процесс OSPF и привяжем сети к зоне 0 (рис. 2.41).

```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ospf
msk-nsandryushin-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01#
```

Рис. 2.41: Настройка OSPF на gw-01

Настроим OSPF на маршрутизаторе gw-02, указав сети и зону (рис. 2.42).



```

msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router ospf
msk-nsandryushin-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-nsandryushin-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █

```

Рис. 2.42: Настройка OSPF на gw-02

Настроим OSPF на маршрутизаторе gw-03 (рис. 2.43).

```

msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ospf
msk-nsandryushin-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# █

```

Рис. 2.43: Настройка OSPF на gw-03

Настроим OSPF на маршрутизаторе gw-04 (рис. 2.44).

```

msk-nsandryushin-gw-04(config-if)# router ospf
msk-nsandryushin-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-nsandryushin-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# █

```

Рис. 2.44: Настройка OSPF на gw-04

Проверим связность с помощью ping и trace с PC1. Трассировка показывает маршрут через gw-02 (рис. 2.45).

```

VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=13.670 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=2.687 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=8.155 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=7.790 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=6.081 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1   5.541 ms  2.604 ms  4.040 ms
 2  10.0.4.1    9.293 ms  3.198 ms  6.435 ms
 3  10.0.2.2   12.825 ms  7.816 ms  6.390 ms
 4  10.0.11.10  14.574 ms  9.895 ms  13.379 ms

VPCS> █

```

Рис. 2.45: Проверка связности через OSPF

Посмотрим список соседей OSPF и таблицу маршрутизации на gw-01 (рис. 2.46).

```

msk-nsandryushin-gw-01# show ip ospf neighbor
Neighbor ID      Pri State          Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup    3m27s            31.783s 10.0.1.2      et
h1:10.0.1.1
10.0.4.1         1 Full/Backup    1m21s            29.990s 10.0.4.1      et
h2:10.0.4.2
0                0              0

msk-nsandryushin-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
    via 10.0.1.2, eth1
N   10.0.3.0/24      [200] area: 0.0.0.0
    via 10.0.4.1, eth2
N   10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.1.2, eth1
    via 10.0.4.1, eth2

===== OSPF router routing table =====
===== OSPF external routing table =====

msk-nsandryushin-gw-01# █

```

Рис. 2.46: Таблица соседей и маршрутов OSPF на gw-01

Повторно выведем информацию о соседях и маршрутах OSPF на gw-01 (рис.

2.47).

```
msk-nsandryushin-gw-01# show ip ospf neighbor

Neighbor ID      Pri State           Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup      4m01s            37.654s 10.0.1.2      et
hl:10.0.1.1
10.0.4.1         1 Full/Backup      1m55s            35.861s 10.0.4.1      et
h2:10.0.4.2

msk-nsandryushin-gw-01# show ip ospf route
===== OSPF network routing table =====
N   10.0.1.0/24      [100] area: 0.0.0.0
    directly attached to eth1
N   10.0.2.0/24      [200] area: 0.0.0.0
    via 10.0.1.2, eth1
N   10.0.3.0/24      [300] area: 0.0.0.0
    via 10.0.1.2, eth1
N   10.0.4.0/24      [100] area: 0.0.0.0
    directly attached to eth2
N   10.0.10.0/24     [100] area: 0.0.0.0
    directly attached to eth0
N   10.0.11.0/24     [300] area: 0.0.0.0
    via 10.0.1.2, eth1

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-nsandryushin-gw-01#
```

Рис. 2.47: Мониторинг OSPF на gw-01

Снова выполним трассировку маршрута с PC1 до PC2 (рис. 2.48).

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.639 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.683 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=4.127 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.546 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=6.524 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    1.486 ms  0.439 ms  0.806 ms
 2  10.0.1.2     6.410 ms  1.842 ms  2.354 ms
 3  10.0.2.2     4.067 ms  2.309 ms  4.865 ms
 4  10.0.11.10   4.043 ms  5.901 ms  6.870 ms

VPCS>
```

Рис. 2.48: Повторная трассировка IPv4

Убедимся, что интерфейс eth0 на gw-04 включен (команда `no shutdown`) (рис. 2.49).

```
msk-nsandryushin-gw-04(config-if)# interface eth0
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 2.49: Включение интерфейса на gw-04

Проверим трассировку маршрута; теперь трафик может идти через gw-04 (рис. 2.50).

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=3.312 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.454 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.039 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.097 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=5.922 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    0.851 ms  1.516 ms  0.552 ms
 2  10.0.4.1     3.025 ms  1.893 ms  1.465 ms
 3  10.0.2.2     2.638 ms  5.644 ms  4.798 ms
 4  10.0.11.10   4.833 ms  6.347 ms  4.860 ms

VPCS>
```

Рис. 2.50: Трассировка через восстановленный маршрут

В Wireshark проанализируем пакеты, среди которых видны сообщения ICMPv6 о недоступности назначения и превышении времени (рис. 2.51).

|    |          |                        |            |        |                                                                      |
|----|----------|------------------------|------------|--------|----------------------------------------------------------------------|
| 1  | 0.000000 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=1, hop limit=64 (reply in 2)  |
| 2  | 0.003380 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=58 (request in 1)  |
| 3  | 1.007774 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=2, hop limit=64 (reply in 4)  |
| 4  | 1.010543 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=58 (request in 3)  |
| 5  | 2.012440 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=3, hop limit=64 (reply in 6)  |
| 6  | 2.016682 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=58 (request in 5)  |
| 7  | 3.020346 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=4, hop limit=64 (reply in 8)  |
| 8  | 3.024414 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=58 (request in 7)  |
| 9  | 4.026622 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x2fdb, seq=5, hop limit=64 (reply in 10) |
| 10 | 4.030371 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=58 (request in 9)  |
| 11 | 5.074644 | fe80::ed5:34ff:fef4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |
| 12 | 5.784419 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 13 | 5.785799 | 2001:10::1             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 14 | 5.787845 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 15 | 5.788581 | 2001:10::1             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 16 | 5.789058 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 17 | 5.789470 | 2001:10::1             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 18 | 5.790696 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 19 | 5.794221 | 2001:11::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 20 | 5.796092 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 21 | 5.798194 | 2001:11::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 22 | 5.800355 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 23 | 5.803391 | 2001:11::2             | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 24 | 5.804815 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 25 | 5.809429 | 2001:2::2              | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 26 | 5.810863 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 27 | 5.816224 | 2001:2::2              | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 28 | 5.817297 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 29 | 5.820823 | 2001:2::2              | 2001:10::a | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                    |
| 30 | 5.821051 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 31 | 5.828793 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 32 | 5.830154 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 33 | 5.835100 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 34 | 5.836864 | 2001:10::a             | 2001:11::a | UDP    | 126 37568 → 37569 Len=64                                             |
| 35 | 5.843583 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]     |
| 36 | 6.113915 | fe80::ed5:34ff:fef4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |
| 37 | 7.154049 | fe80::ed5:34ff:fef4... | 2001:10::a | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00       |

Рис. 2.51: Анализ ошибок ICMPv6 в Wireshark

Продолжаем наблюдение за служебным трафиком OSPF Hello и RIPv2 в Wireshark (рис. 2.52).

|    |            |                        |                  |        |                                                                   |
|----|------------|------------------------|------------------|--------|-------------------------------------------------------------------|
| 52 | 195.884674 | fe80::ed5:34ff:fef4... | ff02::1          | ICMPv6 | 110 Router Advertisement from 0c:d5:34:f4:00:00                   |
| 53 | 199.661651 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 54 | 226.920117 | 10.0.10.1              | 224.0.0.22       | IGMPv3 | 60 Membership Report / Join group 224.0.0.5 for any sources       |
| 55 | 226.921262 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 56 | 227.192342 | 10.0.10.1              | 224.0.0.22       | IGMPv3 | 60 Membership Report / Join group 224.0.0.5 for any sources       |
| 57 | 229.519281 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 58 | 234.664902 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 59 | 236.920946 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 60 | 246.922178 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 61 | 256.924322 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 62 | 263.672737 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 63 | 266.925478 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 64 | 266.933389 | 10.0.10.1              | 224.0.0.22       | IGMPv3 | 60 Membership Report / Join group 224.0.0.6 for any sources       |
| 65 | 267.994423 | 10.0.10.1              | 224.0.0.22       | IGMPv3 | 60 Membership Report / Join group 224.0.0.6 for any sources       |
| 66 | 272.528165 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 67 | 276.927290 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 68 | 286.928528 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 69 | 294.684427 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 70 | 296.931992 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 71 | 306.935385 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 72 | 315.540811 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 73 | 316.939135 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 74 | 325.692372 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 75 | 326.940765 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 76 | 336.943400 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 77 | 346.947587 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 78 | 348.697735 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 79 | 353.549690 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 80 | 356.950800 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 81 | 366.952326 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 82 | 376.707171 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 83 | 376.954518 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 84 | 386.956580 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 85 | 388.559679 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 86 | 396.960370 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 87 | 406.719743 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 88 | 406.962616 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 89 | 416.966892 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 90 | 426.969836 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 91 | 431.575248 | fe80::ed5:34ff:fef4... | ff02::9          | RIPng  | 186 Command Response, Version 1                                   |
| 92 | 436.972058 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 93 | 437.729903 | 10.0.10.1              | 224.0.0.9        | RIPv2  | 146 Response                                                      |
| 94 | 446.974797 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 95 | 456.977776 | 10.0.10.1              | 224.0.0.5        | OSPF   | 78 Hello Packet                                                   |
| 96 | 463.255135 | Private_66:68:00       | Broadcast        | ARP    | 64 Who has 10.0.10.1? Tell 10.0.10.10                             |
| 97 | 463.256583 | 0c:d5:34:f4:00:00      | Private_66:68:00 | ARP    | 60 10.0.10.1 is at 0c:d5:34:f4:00:00                              |
| 98 | 463.258102 | 10.0.10.10             | 10.0.11.10       | ICMP   | 98 Echo (ping) request id=0xfedc, seq=1/256, ttl=64 (reply in 99) |

Рис. 2.52: Анализ OSPF и RIP трафика

Зафиксируем ICMP-запросы и ARP-пакеты в Wireshark (рис. 2.53).

|     |            |                              |                   |       |     |                                                                             |
|-----|------------|------------------------------|-------------------|-------|-----|-----------------------------------------------------------------------------|
| 97  | 463.256583 | 0c:d5:34:f4:00:00            | Private_66:68:00  | ARP   | 60  | 10.0.10.1 is at 0c:d5:34:f4:00:00                                           |
| 98  | 463.258102 | 10.0.10.10                   | 10.0.11.10        | ICMP  | 98  | Echo (ping) request id=0xfedc, seq=1/256, ttl=64 (reply in 99)              |
| 99  | 463.271539 | 10.0.11.10                   | 10.0.10.10        | ICMP  | 98  | Echo (ping) reply id=0xfedc, seq=1/256, ttl=61 (request in 98)              |
| 100 | 464.273975 | 10.0.10.10                   | 10.0.11.10        | ICMP  | 98  | Echo (ping) request id=0xffdc, seq=2/512, ttl=64 (reply in 101)             |
| 101 | 464.276479 | 10.0.11.10                   | 10.0.10.10        | ICMP  | 98  | Echo (ping) reply id=0xffdc, seq=2/512, ttl=61 (request in 100)             |
| 102 | 465.278281 | 10.0.10.10                   | 10.0.11.10        | ICMP  | 98  | Echo (ping) request id=0x00dd, seq=3/768, ttl=64 (reply in 103)             |
| 103 | 465.285973 | 10.0.11.10                   | 10.0.10.10        | ICMP  | 98  | Echo (ping) reply id=0x00dd, seq=3/768, ttl=61 (request in 102)             |
| 104 | 466.287283 | 10.0.10.10                   | 10.0.11.10        | ICMP  | 98  | Echo (ping) request id=0x01dd, seq=4/1024, ttl=64 (reply in 105)            |
| 105 | 466.294795 | 10.0.11.10                   | 10.0.10.10        | ICMP  | 98  | Echo (ping) reply id=0x01dd, seq=4/1024, ttl=61 (request in 104)            |
| 106 | 466.738036 | 10.0.10.1                    | 224.0.0.9         | RIPv2 | 146 | Response                                                                    |
| 107 | 466.981035 | 10.0.10.1                    | 224.0.0.5         | OSPF  | 78  | Hello Packet                                                                |
| 108 | 467.297318 | 10.0.10.10                   | 10.0.11.10        | ICMP  | 98  | Echo (ping) request id=0x02dd, seq=5/1280, ttl=64 (reply in 109)            |
| 109 | 467.302937 | 10.0.11.10                   | 10.0.10.10        | ICMP  | 98  | Echo (ping) reply id=0x02dd, seq=5/1280, ttl=61 (request in 108)            |
| 110 | 468.349414 | 0c:d5:34:f4:00:00            | Private_66:68:00  | ARP   | 60  | Who has 10.0.10.10? Tell 10.0.10.1                                          |
| 111 | 468.349737 | Private_66:68:00             | 0c:d5:34:f4:00:00 | ARP   | 60  | 10.0.10.10 is at 00:50:79:66:68:00                                          |
| 112 | 469.879162 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251781 TSecr=0 WS... |
| 113 | 469.884142 | 10.0.10.1                    | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 114 | 469.884691 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 115 | 469.886865 | 10.0.10.1                    | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 116 | 469.888962 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 117 | 469.892378 | 10.0.10.1                    | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 118 | 469.894329 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 119 | 469.903212 | 10.0.4.1                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 120 | 469.903740 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 121 | 469.906473 | 10.0.4.1                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 122 | 469.909118 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 123 | 469.914524 | 10.0.4.1                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 124 | 469.917170 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 125 | 469.929172 | 10.0.2.2                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 126 | 469.930011 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 127 | 469.937228 | 10.0.2.2                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 128 | 469.937683 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 129 | 469.943748 | 10.0.2.2                     | 10.0.10.10        | ICMP  | 102 | Time-to-live exceeded (Time to live exceeded in transit)                    |
| 130 | 469.945709 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 131 | 469.959532 | 10.0.11.10                   | 10.0.10.10        | TCP   | 60  | 6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0                              |
| 132 | 469.960436 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | [TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=... |
| 133 | 469.970051 | 10.0.11.10                   | 10.0.10.10        | TCP   | 60  | [TCP Port numbers reused] 6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0    |
| 134 | 469.971020 | 10.0.10.10                   | 10.0.11.10        | TCP   | 74  | 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251781 TSecr=0 WS... |
| 135 | 469.984117 | 10.0.11.10                   | 10.0.10.10        | TCP   | 60  | [TCP Port numbers reused] 6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0    |
| 136 | 475.590429 | fe80::ed5:34ff:fef4::ff02::9 |                   | RIPng | 186 | Command Response, Version 1                                                 |
| 137 | 476.985317 | 10.0.10.1                    | 224.0.0.5         | OSPF  | 78  | Hello Packet                                                                |
| 138 | 486.988012 | 10.0.10.1                    | 224.0.0.5         | OSPF  | 78  | Hello Packet                                                                |

Рис. 2.53: Анализ ICMP и ARP

Наблюдаем TCP-ретрансмиссии и сообщения об истечении времени жизни пакета (Time-to-live exceeded) в Wireshark (рис. 2.54).



|     |            |                       |                   |       |                                                                                |                                                |
|-----|------------|-----------------------|-------------------|-------|--------------------------------------------------------------------------------|------------------------------------------------|
| 159 | 580.207362 | 10.0.11.10            | 10.0.10.10        | ICMP  | 98 Echo (ping) reply                                                           | id=0x75dd, seq=1/420, ttl=61 (request in 150)  |
| 160 | 581.271187 | 10.0.10.10            | 10.0.11.10        | ICMP  | 98 Echo (ping) request                                                         | id=0x74dd, seq=2/512, ttl=64 (reply in 161)    |
| 161 | 581.277495 | 10.0.11.10            | 10.0.10.10        | ICMP  | 98 Echo (ping) reply                                                           | id=0x74dd, seq=2/512, ttl=61 (request in 160)  |
| 162 | 581.783442 | 10.0.10.1             | 224.0.0.9         | RIPv2 | 146 Response                                                                   |                                                |
| 163 | 582.280220 | 10.0.10.10            | 10.0.11.10        | ICMP  | 98 Echo (ping) request                                                         | id=0x75dd, seq=3/768, ttl=64 (reply in 164)    |
| 164 | 582.284076 | 10.0.11.10            | 10.0.10.10        | ICMP  | 98 Echo (ping) reply                                                           | id=0x75dd, seq=3/768, ttl=61 (request in 163)  |
| 165 | 583.284995 | 10.0.10.10            | 10.0.11.10        | ICMP  | 98 Echo (ping) request                                                         | id=0x76dd, seq=4/1024, ttl=64 (reply in 166)   |
| 166 | 583.288238 | 10.0.11.10            | 10.0.10.10        | ICMP  | 98 Echo (ping) reply                                                           | id=0x76dd, seq=4/1024, ttl=61 (request in 165) |
| 167 | 584.289819 | 10.0.10.10            | 10.0.11.10        | ICMP  | 98 Echo (ping) request                                                         | id=0x77dd, seq=5/1280, ttl=64 (reply in 168)   |
| 168 | 584.296140 | 10.0.11.10            | 10.0.10.10        | ICMP  | 98 Echo (ping) reply                                                           | id=0x77dd, seq=5/1280, ttl=61 (request in 167) |
| 169 | 585.336635 | 0c:d5:34:f4:00:00     | Private_66:68:00  | ARP   | 60 Who has 10.0.10.10? Tell 10.0.10.1                                          |                                                |
| 170 | 585.336887 | Private_66:68:00      | 0c:d5:34:f4:00:00 | ARP   | 60 10.0.10.10 is at 00:50:79:66:68:00                                          |                                                |
| 171 | 587.000288 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251898 TSecr=0 ... |                                                |
| 172 | 587.001585 | 10.0.10.1             | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 173 | 587.003277 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 174 | 587.003593 | 10.0.10.1             | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 175 | 587.005335 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 176 | 587.006029 | 10.0.10.1             | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 177 | 587.006669 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 178 | 587.012729 | 10.0.1.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 179 | 587.013095 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 180 | 587.014663 | 10.0.1.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 181 | 587.015943 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 182 | 587.018007 | 10.0.1.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 183 | 587.019059 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 184 | 587.022876 | 10.0.2.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 185 | 587.023315 | 10.0.10.1             | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |                                                |
| 186 | 587.024239 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 187 | 587.026426 | 10.0.2.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 188 | 587.028077 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 189 | 587.032320 | 10.0.2.2              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |                                                |
| 190 | 587.034633 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 191 | 587.038276 | 10.0.11.10            | 10.0.10.10        | TCP   | 60 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0                            |                                                |
| 192 | 587.039535 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |                                                |
| 193 | 587.044750 | 10.0.11.10            | 10.0.10.10        | TCP   | 60 [TCP Port numbers reused] 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len... |                                                |
| 194 | 587.045745 | 10.0.10.10            | 10.0.11.10        | TCP   | 74 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251898 TSecr=0 ... |                                                |
| 195 | 587.052134 | 10.0.11.10            | 10.0.10.10        | TCP   | 60 [TCP Port numbers reused] 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len... |                                                |
| 196 | 594.648414 | fe80::ed5:34ff:fe4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |                                                |
| 197 | 597.028063 | 10.0.10.1             | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |                                                |
| 198 | 607.030666 | 10.0.10.1             | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |                                                |
| 199 | 609.798090 | 10.0.10.1             | 224.0.0.9         | RIPv2 | 146 Response                                                                   |                                                |
| 200 | 611.659794 | fe80::ed5:34ff:fe4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |                                                |
| 201 | 617.033846 | 10.0.10.1             | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |                                                |
| 202 | 627.038916 | 10.0.10.1             | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |                                                |

Рис. 2.54: Анализ ошибок TCP и ICMP

Настроим OSPFv3 для IPv6 на маршрутизаторе gw-01, задав router-id 1.1.1.1 и привязав интерфейсы к зоне 0 (рис. 2.55).



|     |            |                        |                   |       |                                                                                |
|-----|------------|------------------------|-------------------|-------|--------------------------------------------------------------------------------|
| 217 | 653.274971 | 10.0.10.10             | 10.0.11.10        | ICMP  | 98 Echo (ping) request id=0xbcd, seq=5/1280, ttl=64 (reply in 218)             |
| 218 | 653.280482 | 10.0.11.10             | 10.0.10.10        | ICMP  | 98 Echo (ping) reply id=0xbcd, seq=5/1280, ttl=61 (request in 217)             |
| 219 | 654.315894 | 0c:d5:34:f4:00:00      | Private_66:68:00  | ARP   | 60 Who has 10.0.10.10? Tell 10.0.10.1                                          |
| 220 | 654.316090 | Private_66:68:00       | 0c:d5:34:f4:00:00 | ARP   | 60 10.0.10.10 is at 00:50:79:66:68:00                                          |
| 221 | 655.640509 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251967 TSecr=0 ... |
| 222 | 655.641074 | 10.0.10.1              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 223 | 655.642653 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 224 | 655.643832 | 10.0.10.1              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 225 | 655.644947 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 226 | 655.645358 | 10.0.10.1              | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 227 | 655.647136 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 228 | 655.649759 | 10.0.4.1               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 229 | 655.651017 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 230 | 655.652586 | 10.0.4.1               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 231 | 655.653749 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 232 | 655.655059 | 10.0.4.1               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 233 | 655.656048 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 234 | 655.658539 | 10.0.2.2               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 235 | 655.659645 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 236 | 655.665087 | 10.0.2.2               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 237 | 655.667343 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 238 | 655.671932 | 10.0.2.2               | 10.0.10.10        | ICMP  | 102 Time-to-live exceeded (Time to live exceeded in transit)                   |
| 239 | 655.673434 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 240 | 655.678080 | 10.0.11.10             | 10.0.10.10        | TCP   | 60 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0                            |
| 241 | 655.678959 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva... |
| 242 | 655.685145 | 10.0.11.10             | 10.0.10.10        | TCP   | 60 [TCP Port numbers reused] 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len... |
| 243 | 655.687478 | 10.0.10.10             | 10.0.11.10        | TCP   | 74 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251967 TSecr=0 ... |
| 244 | 655.692145 | 10.0.11.10             | 10.0.10.10        | TCP   | 60 [TCP Port numbers reused] 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len... |
| 245 | 657.055172 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 246 | 664.823250 | 10.0.10.1              | 224.0.0.9         | RIPV2 | 146 Response                                                                   |
| 247 | 667.058254 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 248 | 669.686874 | fe80::ed5:34ff:fef4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |
| 249 | 677.062567 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 250 | 687.063704 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 251 | 694.832511 | 10.0.10.1              | 224.0.0.9         | RIPV2 | 146 Response                                                                   |
| 252 | 697.065098 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 253 | 698.693610 | fe80::ed5:34ff:fef4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |
| 254 | 707.065061 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 255 | 717.066048 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 256 | 718.695616 | fe80::ed5:34ff:fef4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |
| 257 | 718.836212 | 10.0.10.1              | 224.0.0.9         | RIPV2 | 146 Response                                                                   |
| 258 | 727.066089 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 259 | 734.700750 | fe80::ed5:34ff:fef4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |
| 260 | 737.065635 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 261 | 747.068017 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 262 | 754.839342 | 10.0.10.1              | 224.0.0.9         | RIPV2 | 146 Response                                                                   |
| 263 | 756.708894 | fe80::ed5:34ff:fef4... | ff02::9           | RIPng | 186 Command Response, Version 1                                                |
| 264 | 757.069414 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |
| 265 | 767.069174 | 10.0.10.1              | 224.0.0.5         | OSPF  | 78 Hello Packet                                                                |

Рис. 2.55: Настройка OSPFv3 на gw-01

Настроим OSPFv3 на маршрутизаторе gw-02 с router-id 2.2.2.2 (рис. 2.56).

```

msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ospf6
msk-nsandryushin-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-nsandryushin-gw-01(config-ospf6)# exit
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █

```

Рис. 2.56: Настройка OSPFv3 на gw-02

Настроим OSPFv3 на маршрутизаторе gw-03 с router-id 3.3.3.3 (рис. 2.57).

```

msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router ospf6
msk-nsandryushin-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-nsandryushin-gw-02(config-ospf6)# exit
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █

```

Рис. 2.57: Настройка OSPFv3 на gw-03

Настроим OSPFv3 на маршрутизаторе gw-04 с router-id 4.4.4.4 (рис. 2.58).

```

msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ospf6
msk-nsandryushin-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-nsandryushin-gw-03(config-ospf6)# exit
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# █

```

Рис. 2.58: Настройка OSPFv3 на gw-04

Проверим IPv6-связность с PC1. Трассировка проходит успешно (рис. 2.59).

```

msk-nsandryushin-gw-04(config-if)# router ospf6
msk-nsandryushin-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-nsandryushin-gw-04(config-ospf6)# exit
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# █

```

Рис. 2.59: Проверка связности OSPFv3

Проверим таблицу соседей и маршрутов OSPFv3 на gw-01 (рис. 2.60).

```

VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=28.349 ms
2001:11::a icmp6_seq=2 ttl=58 time=8.277 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.083 ms
2001:11::a icmp6_seq=4 ttl=58 time=9.734 ms
2001:11::a icmp6_seq=5 ttl=58 time=11.764 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1 4.485 ms 0.880 ms 2.130 ms
 2 2001:1::2 4.217 ms 5.025 ms 3.368 ms
 3 2001:2::2 11.396 ms 3.765 ms 3.199 ms
 4 2001:11::a 5.634 ms 5.830 ms 3.506 ms

VPCS> █

```

Рис. 2.60: Проверка OSPFv3 на gw-01

В Wireshark наблюдаем сообщения ICMPv6 Time Exceeded, свидетельствующие о проблемах доставки пакетов (рис. 2.61).

```

msk-nsandryushin-gw-01# show ipv6 ospf6 neighbor
Neighbor ID Pri DeadTime State/IfState Duration I/F[State]
2.2.2.2 1 00:00:38 Full/BDR 00:03:07 eth1[DR]
4.4.4.4 1 00:00:38 Full/BDR 00:01:18 eth2[DR]
msk-nsandryushin-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64 :: eth1 00:01:21
*N IA 2001:2::/64 fe80::ed8:daff:fe5f:0 eth1 00:01:21
*N IA 2001:3::/64 fe80::e05:bfff:feca:1 eth2 00:01:21
*N IA 2001:4::/64 :: eth2 00:01:21
*N IA 2001:10::/64 :: eth0 00:01:21
*N IA 2001:11::/64 fe80::ed8:daff:fe5f:0 eth1 00:01:21
fe80::e05:bfff:feca:1 eth2
msk-nsandryushin-gw-01# █

```

Рис. 2.61: Анализ ошибок Time Exceeded в Wireshark

Фиксируем сообщения ICMPv6 Destination Unreachable (Port unreachable) в трафике (рис. 2.62).

|     |             |                        |                        |        |                                                                         |
|-----|-------------|------------------------|------------------------|--------|-------------------------------------------------------------------------|
| 358 | 1112.140491 | fe80::ed5:34ff:fef4... | ff02::5                | OSPF   | 90 Hello Packet                                                         |
| 359 | 1113.144336 | 2001:10::a             | 2001:11::a             | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=1, hop limit=64 (reply in 362)   |
| 360 | 1113.171209 | fe80::ed5:34ff:fef4... | ff02::1:ff00:a         | ICMPv6 | 86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00          |
| 361 | 1113.171474 | 2001:10::a             | fe80::ed5:34ff:fef4... | ICMPv6 | 86 Neighbor Advertisement 2001:10::a (sol, ovr) is at 00:50:79:66:68:00 |
| 362 | 1113.172124 | 2001:11::a             | 2001:10::a             | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=1, hop limit=58 (request in 359)   |
| 363 | 1114.174186 | 2001:10::a             | 2001:11::a             | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=2, hop limit=64 (reply in 364)   |
| 364 | 1114.181911 | 2001:11::a             | 2001:10::a             | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=2, hop limit=58 (request in 363)   |
| 365 | 1115.185194 | 2001:10::a             | 2001:11::a             | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=3, hop limit=64 (reply in 366)   |
| 366 | 1115.190019 | 2001:11::a             | 2001:10::a             | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=3, hop limit=58 (request in 365)   |
| 367 | 1116.192889 | 2001:10::a             | 2001:11::a             | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=4, hop limit=64 (reply in 368)   |
| 368 | 1116.202029 | 2001:11::a             | 2001:10::a             | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=4, hop limit=58 (request in 367)   |
| 369 | 1116.957509 | 10.0.10.1              | 224.0.0.9              | RIPv2  | 146 Response                                                            |
| 370 | 1117.150945 | 10.0.10.1              | 224.0.0.5              | OSPF   | 78 Hello Packet                                                         |
| 371 | 1117.203386 | 2001:10::a             | 2001:11::a             | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=5, hop limit=64 (reply in 372)   |
| 372 | 1117.211232 | 2001:11::a             | 2001:10::a             | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=5, hop limit=58 (request in 371)   |
| 373 | 1118.760537 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 374 | 1118.764676 | 2001:10::1             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 375 | 1118.766478 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 376 | 1118.767117 | 2001:10::1             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 377 | 1118.768781 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 378 | 1118.770136 | 2001:10::1             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 379 | 1118.771971 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 380 | 1118.775962 | 2001:11::2             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 381 | 1118.777337 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 382 | 1118.781946 | 2001:11::2             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 383 | 1118.783628 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 384 | 1118.786701 | 2001:11::2             | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 385 | 1118.789506 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 386 | 1118.800382 | 2001:2::2              | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 387 | 1118.800954 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 388 | 1118.804550 | 2001:2::2              | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 389 | 1118.806643 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 390 | 1118.809646 | 2001:2::2              | 2001:10::a             | ICMPv6 | 174 Time Exceeded (Hop limit exceeded in transit)                       |
| 391 | 1118.811238 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 392 | 1118.816679 | 2001:11::a             | 2001:10::a             | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]        |
| 393 | 1118.817280 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 394 | 1118.822895 | 2001:11::a             | 2001:10::a             | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]        |
| 395 | 1118.824143 | 2001:10::a             | 2001:11::a             | UDP    | 126 17184 → 17185 Len=64                                                |
| 396 | 1118.827416 | 2001:11::a             | 2001:10::a             | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]        |
| 397 | 1122.144008 | fe80::ed5:34ff:fef4... | ff02::5                | OSPF   | 90 Hello Packet                                                         |
| 398 | 1127.153389 | 10.0.10.1              | 224.0.0.5              | OSPF   | 78 Hello Packet                                                         |
| 399 | 1132.147446 | fe80::ed5:34ff:fef4... | ff02::5                | OSPF   | 90 Hello Packet                                                         |
| 400 | 1137.157307 | 10.0.10.1              | 224.0.0.5              | OSPF   | 78 Hello Packet                                                         |
| 401 | 1139.973818 | 10.0.10.1              | 224.0.0.9              | RIPv2  | 146 Response                                                            |
| 402 | 1142.150435 | fe80::ed5:34ff:fef4... | ff02::5                | OSPF   | 90 Hello Packet                                                         |
| 403 | 1147.160002 | 10.0.10.1              | 224.0.0.5              | OSPF   | 78 Hello Packet                                                         |
| 404 | 1147.849347 | fe80::ed5:34ff:fef4... | ff02::9                | RIPng  | 186 Command Response, Version 1                                         |

Рис. 2.62: Анализ ошибок Destination Unreachable

Продолжаем наблюдать поток ошибок Destination Unreachable в Wireshark (рис. 2.63).

|     |             |                        |            |        |                                                                       |
|-----|-------------|------------------------|------------|--------|-----------------------------------------------------------------------|
| 254 | 1120.311087 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=2, hop limit=61 (reply in 255) |
| 255 | 1120.311508 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=2, hop limit=61 (request in 254) |
| 256 | 1121.321192 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=3, hop limit=61 (reply in 257) |
| 257 | 1121.321635 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=3, hop limit=61 (request in 256) |
| 258 | 1122.333098 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=4, hop limit=61 (reply in 259) |
| 259 | 1122.333452 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=4, hop limit=61 (request in 258) |
| 260 | 1122.434537 | 10.0.11.1              | 224.0.0.5  | OSPF   | 78 Hello Packet                                                       |
| 261 | 1123.339941 | 2001:10::a             | 2001:11::a | ICMPv6 | 118 Echo (ping) request id=0x88df, seq=5, hop limit=61 (reply in 262) |
| 262 | 1123.340735 | 2001:11::a             | 2001:10::a | ICMPv6 | 118 Echo (ping) reply id=0x88df, seq=5, hop limit=61 (request in 261) |
| 263 | 1123.438047 | fe80::eb5:37ff:fe3f... | ff02::9    | RIPng  | 186 Command Response, Version 1                                       |
| 264 | 1124.947469 | 2001:10::a             | 2001:11::a | UDP    | 126 17184 → 17185 Len=64                                              |
| 265 | 1124.948248 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]      |
| 266 | 1124.954346 | 2001:10::a             | 2001:11::a | UDP    | 126 17184 → 17185 Len=64                                              |
| 267 | 1124.954539 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]      |
| 268 | 1124.959105 | 2001:10::a             | 2001:11::a | UDP    | 126 17184 → 17185 Len=64                                              |
| 269 | 1124.959305 | 2001:11::a             | 2001:10::a | ICMPv6 | 174 Destination Unreachable (Port unreachable)[Malformed Packet]      |
| 270 | 1129.977196 | fe80::eb5:37ff:fe3f... | ff02::5    | OSPF   | 90 Hello Packet                                                       |
| 271 | 1132.438347 | 10.0.11.1              | 224.0.0.5  | OSPF   | 78 Hello Packet                                                       |
| 272 | 1133.385374 | 10.0.11.1              | 224.0.0.9  | RIPv2  | 146 Response                                                          |
| 273 | 1139.981233 | fe80::eb5:37ff:fe3f... | ff02::5    | OSPF   | 90 Hello Packet                                                       |
| 274 | 1142.443023 | 10.0.11.1              | 224.0.0.5  | OSPF   | 78 Hello Packet                                                       |
| 275 | 1149.986902 | fe80::eb5:37ff:fe3f... | ff02::5    | OSPF   | 90 Hello Packet                                                       |
| 276 | 1152.446643 | 10.0.11.1              | 224.0.0.5  | OSPF   | 78 Hello Packet                                                       |

Рис. 2.63: Продолжение анализа ошибок в Wireshark

Создадим новый проект «lab8-1» для выполнения следующей части задания (рис. 2.64).

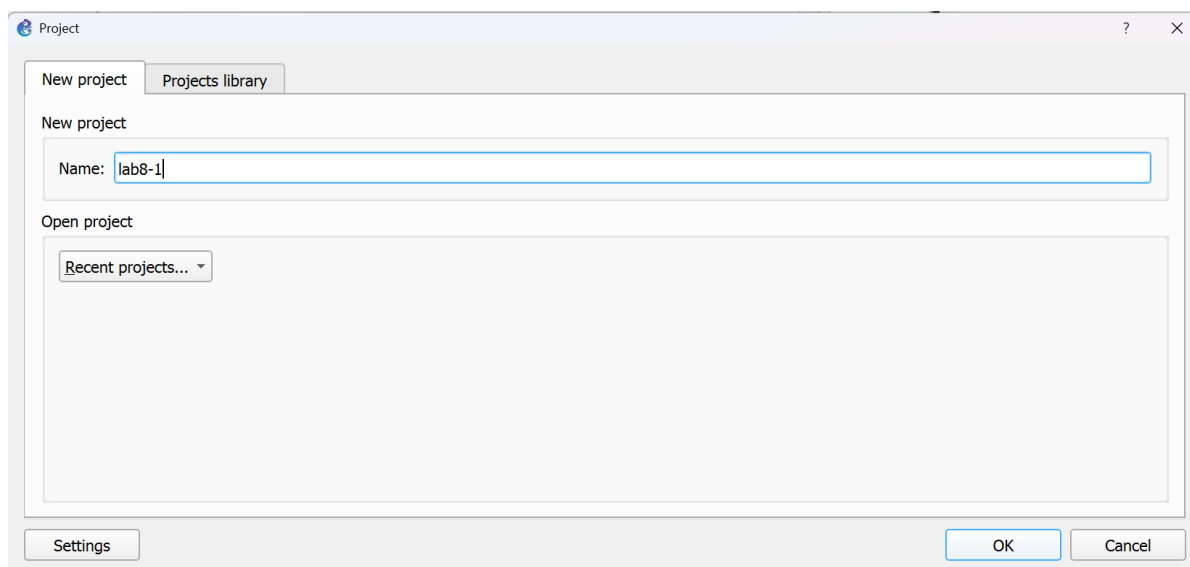


Рис. 2.64: Создание проекта lab8-1

Соберем новую топологию сети с использованием маршрутизаторов VyOS (рис. 2.65).

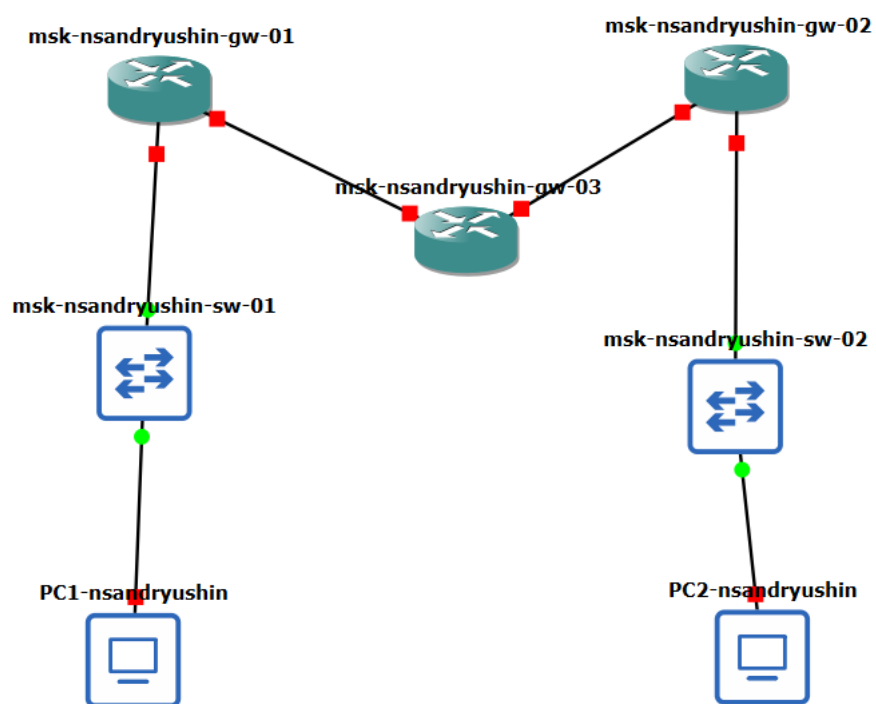


Рис. 2.65: Топология с VyOS

## **3 Выводы**

В результате выполнения лабораторной работы были получены навыки настройки маршрутизации